

Eric Boittier, Ph.D.



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💬 EN (native) DE (B2)

Scientific Software – Deep Learning

Professional Experience

- 2026 • **Postdoctoral Scientist** @Uni. Basel – Basel, CH
 - Machine learning (ML) on graph-structured data
- 2025 • **Teaching Assistant & PhD Researcher**
 - Designed, trained, validated deep learning models
 - 1st place team, HYDRA Spectroscopy Challenge
- 2020 •
 - Peer reviewer, Journal of Chemical Physics
- 2020 • **Research Assistant** @TRI – Brisbane, AU
 - Small-molecule & (bio)molecular modelling, Schrödinger software suite, OpenMM, High Performance Computing
 - Collaborated with multidisciplinary experts, building end-to-end solutions in Python and R
 - Co-supervised and authored research on structure- and ligand-based drug design
- 2019 •

Education

- 2025 • **Doctor of Philosophy, Chemistry** @Uni. Basel – Basel, CH
 - Thesis: Molecular Deep Learning for Quantitative Simulations and Electrostatic Models
- 2020 • **Bachelor of Advanced Science, Chemistry** @Uni. Queensland – Brisbane, AU
 - Thesis: Development of Computational Tools for the Rational Design of Glycosaminoglycan Mimetics.
- 2015 • Honours (1st class), Dean's Commendation

Open Source Software Development



Community-driven/open-source contributions available on GitHub (@EricBoittier)

- **Cheminformatics software** (RDKit) bug fixes for C++ and python bindings, attended User Group Meeting (Zurich, 2024)
- AI for a popular online strategy game using **Reinforcement learning** (Stable Baselines3) with self-play and masked Proximal Policy Optimization
- **Computer vision tool** (OpenCV) for item tracking and **inventory management** (SQL) for an online retailer (CardMarket)
- **Open Source Computational Pipeline** (mmml) for Density Functional Theory and Molecular Dynamics with machine learned interaction potentials written in JAX (python)

Skills & Competencies

Software	Python, C++, Julia, Fortran
Web	Typescript, React, Tailwind CSS
DevOps	Git, CI/CD, Docker
Machine Learning	JAX, PyTorch, scikit-learn
MLOps	Unsloth, Hugging-Face, Litellm, Open-Claw
Data Science	SQL, Polars, Pandas

Invited Talks

- 2025 **Mixing Machine Learned with Empirical Energy Functions**
apoCHARMM Meeting
Boston, US (Online)
- 2024 **Bridging Machine Learning and Electrostatic Models**
Swiss Chemical Society Machine Learning Group
Lausanne, CH